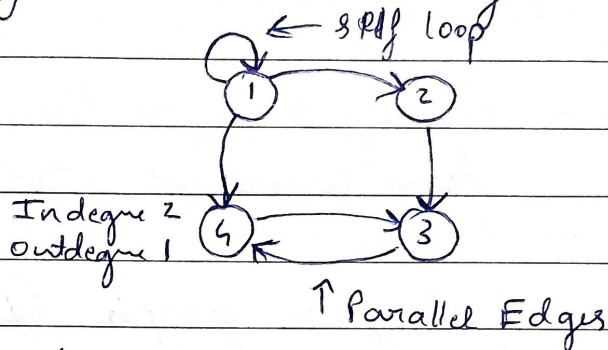


graphs

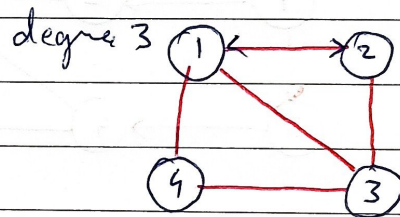
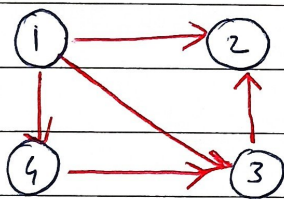
Collection of vertices and edges.

$$G = (V, E)$$



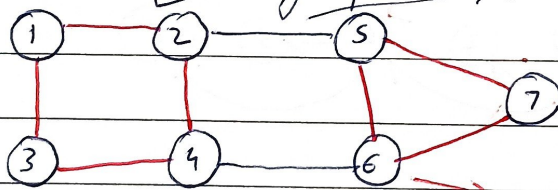
Simple Digraph

non-Directed graph / graph



non-Connected graphs

Biconnected.

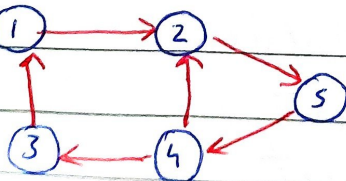


2 components

articulation Points

Strongly Connected.

Directed graph

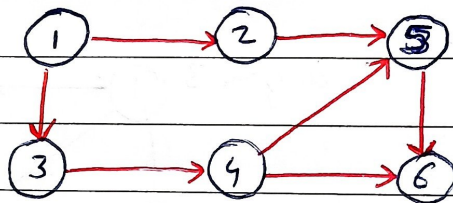


From any vertex we can reach any other vertex

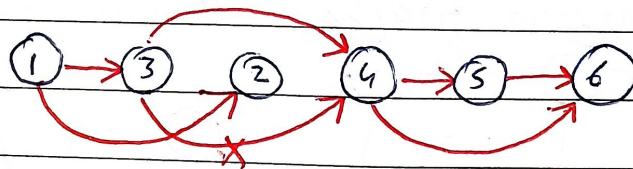
Path - Set of all the vertices in b/w the pair of vertex

Cycle - Coming back to the same vertex from where we started

Directed Acyclic graph (DAG)
(No cycle)



Topological ordering



Representing Graphs

3 ways to represent.

- ① Adjacency Matrix.
- ② Adjacency List
- ③ Compact List.

Adjacency Matrix

	1	2	3	4	5
1	0	1	1	1	0
2	1	0	1	0	0
3	1	1	0	1	1
4	1	0	1	0	1
5	0	0	1	1	0

5x5

represent

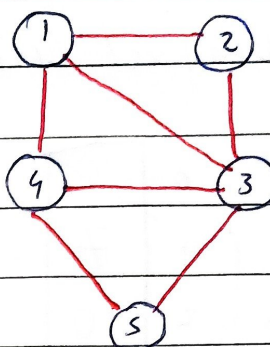
self loop if edge (i,j)

$$A[i][j] = 1$$

or 0

n^2

$$O(n^2)$$

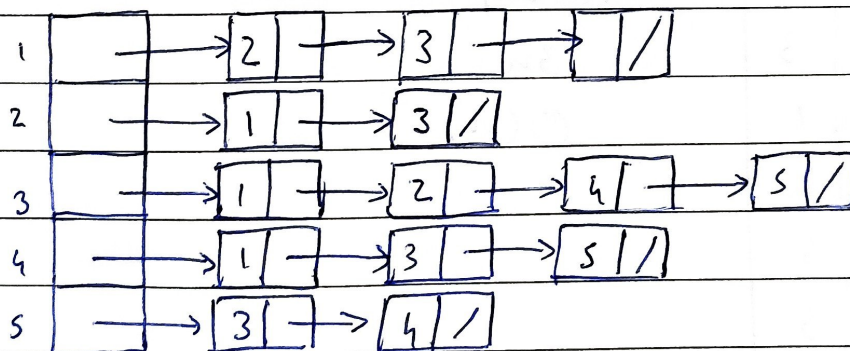


$$G = (V, E)$$

$$|V| = 5$$

$$|E| = 7$$

Adjacency List



$$|V| + 2|E|$$

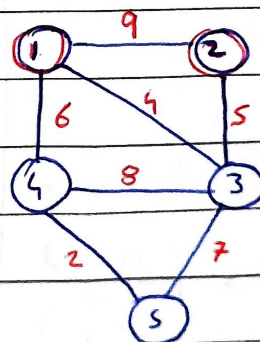
$$n + 2e$$

$$O(n + 2e)$$

	1	2	3	4	5
1	0	9	4	6	0
2	9	0	5	0	0
3	4	5	0	8	7
4	6	0	8	0	2
5	0	0	7	2	0

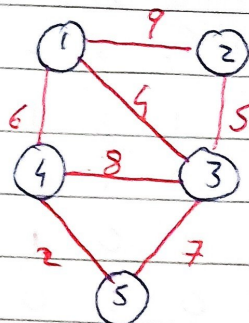
Cost Adjacency Matrix

weight of edges =
cost of edge



Compact List

Vertices		0
1	1	7
	2	10
	3	12
	4	16
	5	19
2	6	21
	7	2
	8	3
3	9	4
	10	1
	11	3
	12	1
4	13	2
	14	4
	15	5
	16	1
5	17	3
	18	5
	19	3
	20	4
	21	



$$G(V, E)$$

$$|V| = n = 5$$

$$|E| = c = 7$$

$$|V| + 2|E| + 1$$

$$5 + 2(7) + 1 = 20$$

$$20 + 1$$

$$n + 2e$$

$$\text{if } c = n$$

$$n + 2n$$

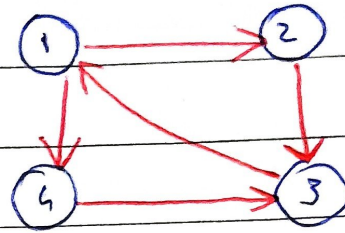
$$3n$$

$$O(n)$$

Adjacency Matrix

	1	2	3	4
1	0	1	0	1
2	0	0	1	0
3	1	0	0	0
4	0	0	1	0

$O(n^2)$

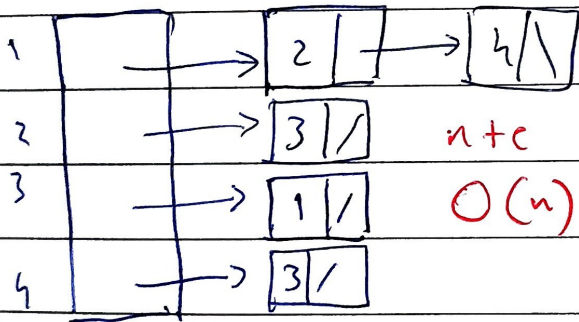


$$G = (V, E)$$

$$|V| = 4$$

$$|E| = 5$$

Adjacency List



$n+1$

$O(n)$

inverse Adjacency List

